

GAS PROCESSING & METHANOL COMPLEX PROJECT NO.11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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HAZID Report

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Revision History

Rev No.	Date	Description of Revision
R01	14-07-22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-RPT-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design

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Abbreviations

Abbreviation	Meaning
ALARP	As Low As Reasonably Practicable
DEP	Design and Engineering Practice
F & G	Fire & Gas
HAC	Hazardous Area Classification
HAZID	Hazard Identification
MAH	Major Accident Hazards
PEFS	Process Engineering Flow scheme
PFS	Process Flow scheme
RAM	Risk Assessment Matrix

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity Greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35&36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: BubouweBou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

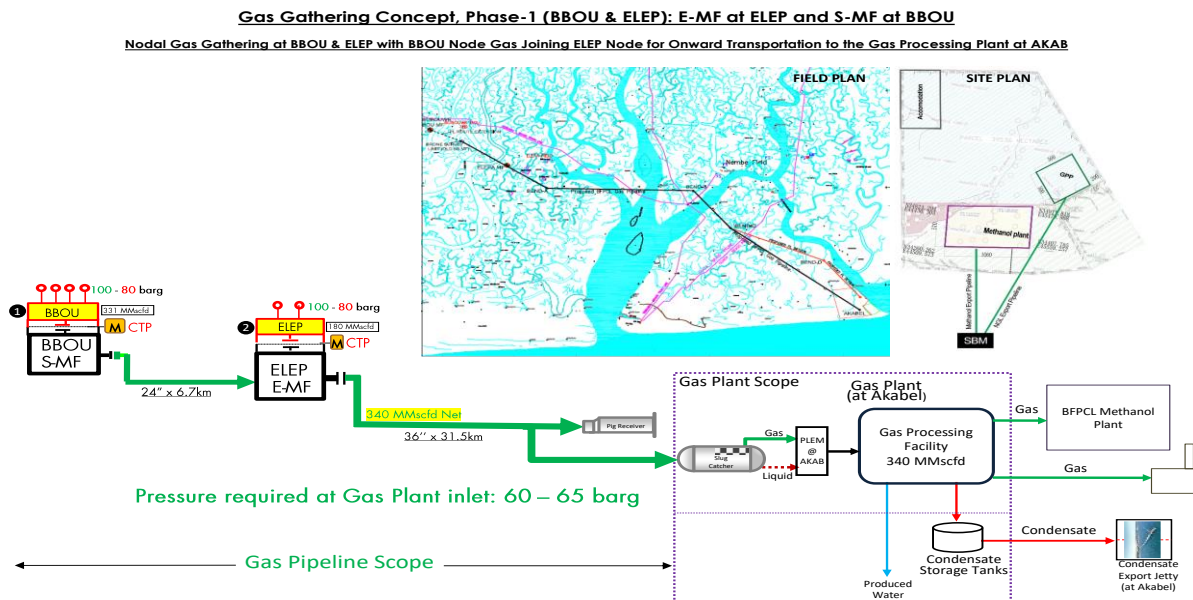


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The objective of the HAZID (Hazard Identification) is to identify the hazards applicable to the FRP project in the detailed engineering design phase and carry out a risk assessment rating, address the control and recovery preparedness measures and make recommendations as appropriate

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.00.00.15-Gen	Hazard identification (HAZID) Study

The latest versions of the regulations, codes, and standards and extant amendments or supplements were applied. Local laws regulating the oil and gas industry in Nigeria was strictly adhered to. The order of precedence for the documents were as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent was applied except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations were referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Methodology

The study method was a combination of identification, analysis, and brainstorming on the hazards identified within the project by breaking down the project into various nodes and “following the molecules”. The identified hazards were assigned a risk ranking. Risk mitigation plans were identified and recommendations were made where the controls were not considered sufficient

The output of the HAZID Study was a record of the discussion detailing the identified hazards, Project Hazard and Effect Register, and recommendations for any risk mitigation actions required for the work covered under the scope

The basic approach involved:

- Assembly of an appropriate team; design team engineers, construction engineers, HSE reps, asset team representatives, in a workshop
- Short presentations by the project team detailing the designs and systems, and construction activities under the scope of the HAZID Study
- Identify hazards related to the “molecules”
- Each of the hazards identified was ranked for severity, the likelihood of occurrence, and consequences on people, assets, environment, and community using the NGC Risk Assessment Matrix.
- Carry out hazard analysis for all the hazards /scenarios and bowtie analysis for the major accident hazards (MAHs)
- Record the discussions on worksheets (Hazard & Effect Register) summarizing the nature of the hazard, its consequences, the current safeguards in place, and recommendations for any actions required.

3.1. Guidewords

HAZID guidewords applicable to the phase were adopted for the review session. The guidewords are intended to prompt discussion and are not intended to be exhaustive. The guidewords are shown in Appendix 1 below.

3.2. Preparation Work, Nodes, and Documentation

A preliminary note delineation was carried out by the Facilitator (ESLP Consultants Limited). The nodes of study are in Table 1 below

Table 1: Proposed HAZID Nodes

Node	Title
1	BBOU Manifold
2	ELEPA Manifold
3	AKABEL Pipeline End Facilities

3.3. Preparation Work, Nodes, and Documentation

The following key documentation was required for the HAZID study

- Facility layout/plot plan incl. major equipment layout and occupied buildings
- Findings from previous HAZID exercise
- Emergency response plans
- Plans for construction, transportation, installation and commissioning
- Neighbouring facilities
- Approved for HAZOP Process Flow Diagram(s)
- Approved for HAZOP Process & Instrumentation Drawings (P & IDs)
- Shell Risk Assessment Matrix (RAM)

3.4. HAZID Agenda

The agenda was as follows:

- Team introduction, domestic arrangements, safety instructions by Client
- HAZID Chairman (Facilitator) introduced the scope of the HAZID study
- Project Team presented the scope of the project, execution strategies, highlighting the key features and decisions taken to date
- The HAZID Chairman explained the HAZID process including the HAZID worksheet format and terminologies
- The Chairman:
 - Proposed Nodes and agree on Nodes with the team.
 - Commenced HAZID using the guidewords.
 - All relevant information was recorded on the worksheet.
 - Comfort breaks were held at intervals as agreed by the team

3.5. HAZID Session Details

The HAZID was carried out on the 7th of July 2022. The session was performed for 1 day (estimated) commenced at 09.00AM and finished by 4:00PM. Attendees/ reps are shown in Table 2 below.

Table 2: HAZID Attendee list

Discipline	Name	Company
ESLP Consultants Reps (HAZID Facilitator)		
HAZOP Chairman	Emeka Maduabuchi	ESLP Consultants Ltd
HAZOP Scribe	Valentine Orafu	ESLP Consultants Ltd
Operations Safety	Ekene Okoye	ESLP Consultant Ltd
BRASS FERTILIZER COMPANY Reps (Project Owner)		
GM, Midstream/Upstream Integration	Ralph Afolabi	BFPCL
Chief Technical Officer	Jaspal Singh	BFPCL
TCE (Project Management Consultant of BFPCL)		
Project Management	Arshi Bhutani	TCE
Project Management	Atul Jain	TCE
Project Management	Rajesh Poojari	TCE
Project Management	Sandeep Mithailal Yadav	TCE
Project Management	Surinder Singh Madan	TCE
Project Management	S Narayanan	TCE
EPIC ATLANTIC Reps (FEED Contractor)		
Project Director/Process	Aloy Duru	EPIC ATLANTIC
Project Management	Maduka Ekpunobi	EPIC ATLANTIC
Civil / Structural	William Irem	EPIC ATLANTIC
Project Services	Obima Joe Oragwu	EPIC ATLANTIC
Operations Readiness and Assurance	Olarinde Ige	EPIC ATLANTIC
Pipelines, Mechanical, Piping	Emeka Nwachukwu	EPIC ATLANTIC
Project Services	Shedrack Anigbankpu	EPIC ATLANTIC
Process	Basil Akpan	EPIC ATLANTIC

The HAZID was recorded in Excel worksheet. The worksheets were projected onto a screen so that team members have an opportunity to review and comment during the session. Actions recorded

during the session was assigned to individual team members for follow-up and closeout after the meeting

3.6. Deliverables & Follow Up Work

The HAZID worksheet presenting a record of the HAZID findings was completed real time during the workshop in the full view of all participants for continuous review and confirmation that the report reflects the session content.

This HAZID report is the deliverable of the HAZID workshop and is issued to the FEED Contractor, with the HAZID worksheet as an attachment to the report, for implementation of the recommendations.

HAZID action tracking sheet is to be maintained by the FEED Contractor until completion of FEED.

3.7. HAZID Results

The HAZID minutes were formally recorded, together with recommendations.

A HAZID worksheet is shown as Appendix 2.

A total of 17 recommendations were made, Table 2.4. Below shows the recommendations

Table 2.4.: HAZID Recommendations

S/N	Recommendations
1.	Obtain data on highest flood in the area and raise the platforms above the highest flood level.
2.	Provide lightening protection and earthing system for all tall structures (vents and water tanks) in the platforms
3.	Develop inclement weather procedure for the operate phase of the project facilities
4.	Confirm shore protection strategy at Akabel and design the base foundation of the facilities at Akabel to suit
5.	Sign off project security plan
6.	Finalize and sign-off proposal to relocate the communities at the Akabel end of the project facilities
7.	Integrate pig trap drain system at Akabel end with Akabel gas plant closed drain system
8.	Carry out dispersion analysis of the vent systems to determine the hazard distances on the platforms

9.	Carry out a 3D model review during detail design phase of the project facilities
10.	Develop an emergency response plan for the platforms
11.	Develop concurrent construction operation plan during the construction phase of the project
12.	Develop access control procedure during operate phase of the project
13.	Carry out HAZOP study of the project facilities
14.	Develop transportation HSSE plan during construction phase of the project
15.	Carry out Logistic Infrastructure Review and Assessment (LIRA) for the project
16.	Develop chemical handling procedure using SHOC, during detailed design
17.	Develop noise mapping of the project facilities taken into consideration the blowdown valves

Appendix 1: Typical HAZID Guidewords

Section 1: External & Environmental Hazards					
S/N	Category	Hazard/Event	Scenario	Y/N	
1.01	Natural and Environmental Hazards	Climate Extremes	Temperature, waves, wind, dust, flooding, sandstorms, ice, blizzards	Y	Flooding
1.02		Lightening		Y	
1.03		Earthquakes		N	
1.04		Erosion	Ground slide, coastal, riverine	Y	
1.05		Subsidence	Ground structure, foundations	Y	
2.01	Created (Man-made) Hazards	Security Hazards	Internal and external security threats	Y	
2.02		Terrorist Activity	Riots, civil disturbance, strikes, military action, political unrest	Y	
3.01	Effect of the Facility on the Surroundings	Geographical – Infrastructure	Plant location, plant layout, pipeline routing	Y	

3.02		Proximity to Population	Proximity to population	Y	
3.03		Adjacent Land Use	Crop burning, air-fields, accommodation camps	Y	
3.04		Proximity to Transport Corridors	Shipping lanes, air routes, roads, etc	Y	
4.01	Environmental Damage	Continuous Plant Discharges to Air	Flares, vents	N	
4.02		Continuous Plant Discharges to Water	Target/legislative requirements, drainage facilities	Y	
4.03		Continuous Plant Discharges to Soil	Drainage, chemical storage	Y	At Akabel
4.04		Emergency/upset Discharges	Flares, vents, drainage	Y	Vents
Section 2: Facility Hazards					
Code	Category	Hazard/Event	Scenario		
5.01		Maintenance Philosophy	Heavy lifting, access	Y	
5.02		Emergency Response	Isolation, ESD philosophy, blowdown, flaring/venting requirements	Y	
5.03		Concurrent Operations	Production, maintenance requirements	Y	
5.04		Start-up / Shutdown		Y	
6.01	Fire and Explosion Hazards	Sources of Ignition	Electricity, flares, sparks, hot surfaces (mitigation measures include: identify, remove, separate)	Y	Electricity, lightening and access control
6.02		Equipment Layout	Confinement, escalation following release of explosive or flammable fluid; Mitigation measures include reduce degree of confinement, spacing based on	Y	

			consequence assessment, escalation barriers)		
6.03		Fire Protection and Response	Active/passive insulation, fire/gas detection, blow-down/relief system philosophy, fire-fighting facilities	Y	
6.04		Operator Protection	Means of escape, PPE, communications, emergency response, plant evacuation	Y	
7.01	Process Hazards	Asset Integrity Management	Process control failure, structural failure, erosion or corrosion (mitigation measures include: recognise and minimise process hazards during design, inherently safe plant, containment and recovery measures)	Y	
7.02		Excess/zero Level	Overfill storage tanks and vessels, blowby to downstream vessels	Y	Closed drain tanks and open drain tank
8.01	Utility Systems	Firewater Systems		N	
8.02		Fuel Gas		N	
8.03		Heating / Cooling Medium		N	
8.04		Potable Water		Y	Utility water
9.01	Maintenance Hazards	Access Requirements		Y	
9.02		Heavy Lifting Requirements		Y	
9.03		Transport		Y	

10.01	Construction/ Existing Facilities	Tie-ins (shutdown requirements)		N	
10.02		Concurrent Operations		N	
10.03		Skid Dimensions (weight handling/equipment congestion)		Y	Pig traps, chemical injection, containerised packages
Section 3: Others					
11.00	Biological	snakes, reptiles		Y	
12.00	Chemicals	glycol, foam systems, water treatment chemicals		Y	Corrosion inhibitor, Biocides injection and scale inhibitor
13.00	Ergonomics	Human machine interface		Y	
14.00	Noise	Blowdown valves, gas generator		Y	Blowdown valve

Appendix 2: Typical HAZID Worksheet

S/N	HAZARD CATEGORY	HAZARD EVENT	CAUSES	SCENARIOS	CONSEQUENCES	Risk / Priority (H/M/L)	EXISTING MITIGATION	RECOMMENDATIONS
1	NATURAL AND ENVIRONMENTAL HAZARDS							
		Climate Extremes	Flooding	Area is seasonally flood-prone, flood level can go as high as ???	Damage to control equipment/ loss of control Potential for electrocution from submerged electrical equipment	H	Design specification (platform installed above highest flood level)	Obtain data on highest flood in the area and raise the platforms above the highest flood level.

			Lightening	Area is lightening prone	Injury / fatality from lightening strike Ignition source on coincidence loss of containment	H	Earthing systems for piping and platforms	Provide lightening protection and earthing system for all tall structures (vents and water tanks) Develop inclement weather procedure for the operate phase of the project
			Erosion	Possible coastal erosion on the platform foundation at Akabel	Wash off of the platform foundations and potentially leading to LOPC	H		Confirm shore protection strategy at Akabel and design the base foundation of facilities at Akabel to suit
			Subsidence	Akabel area is sand filled	Possibility of subsidence leading to foundation damage and possible LOPC	H	Design specifications	
1.2	CREATED (MAN-MADE) HAZARDS							
			Security Hazards	Project is in the Niger Delta, security prone	Release of security hazard potentially leading to injury/fatality and LOPC	H	Project security plan	Sign off project security plan

			Terrorist Activity	See Security hazards		H		
1.3	EFFECT OF THE FACILITY ON THE SURROUNDINGSs							
		Geographical - Infrastructure	Pipeline routing	New RoW (right of way) encroaching on people's land and means of livelihood	Loss of means of livelihood	M	Compensation (social performance)	
		Proximity to Population	Proximity to Population	Pipeline runs through some settlements/communities	Potential human impact in the event of LOPC	H	Design specifications Accounting for settlements Relocation of communities at Akabel end	Finalize and sign-off proposal to relocate the communities at the Akabel end of the project facilities
		Proximity to Transport Corridor	Water transport corridors	There are three major rivers crossing and about ten minor rivers crossing	Potential injury hazards to water transport users	M	Design specification (horizontal directional drilling at all river crossings)	
1.4	ENVIRONMENTAL DAMAGE							

		Continuous Plant Discharges to Water	Drainage from pig traps	Potential discharge of hydrocarbon to the environment during pigging operations	Environmental pollution	M	Closed drain tank provided	
		Continuous Plant Discharges to Soil	Drainage from pig traps at Akabel	Potential discharge of hydrocarbon to the environment during pigging operations	Environmental pollution	M		Integrate pig trap drain system at the Akabel end of the facilities with Akabel gas plant closed drain system
		Emergency/upset Discharges	Vent	Buildup of gas cloud at the vent	Buildup of gas cloud, potential for injury/fatality if ignited Greenhouse gas emission	M		Carry out dispersion analysis of the vents to determine the hazard distances
2	FACILITY HAZARDS							
		Maintenance Philosophy	Heavy lifting, access	Lack of maintenance access Lifting for maintenance	Safety system in jeopardy Dropped object from lifting potentially leading to injury/fatality	M	Jib crane provided for lifting items from boat landing into the platform Lifting of heavy maintenance items to be done by	Carry out a 3D model review of the project facilities during detail design phase of the project

							maintenance contractor	
		Emergency Response	Emergency response	Lack of adequate emergency response	Potential escalation of emergency scenarios	M	Emergency response philosophy	Develop an emergency response plan for the project facilities
		Concurrent Operations		Concurrent construction operations	Potential loss of control possibly leading to injury/fatality	M		Develop concurrent construction operation plan during the construction phase of the project
		Start-up / Shutdown		Potential start-up and shutdown issues	Potential loss of control possibly leading process upset, product quality issues emergency/ upset discharges	M	Start-up and shut-down procedure	
2.2	FIRE AND EXPLOSION HAZARDS							
		Sources of Ignition	Electricity, lightning	Ignition hazards on coincident LOPC	Injury/fatality from ignited release	M	Hazardous Area Classification (HAC)	Develop access control procedure during operate phase of the project

			and personnel					
		Equipment Layout		Escalation of process safety event	Injury/fatality from ignited release	M	Layout optimization	
		Fire Protection and Response		Fire from LOPC scenarios	Injury/fatality from ignited release	H	Layout optimization, Gas detection system tied to ESD/ Blowdown	
		Operator Protection		Fire risk on operators (attended operation)	Injury/fatality from ignited release	M	Emergency response Layout optimization Fire wall between hydrocarbon facility and office	
2.3	PROCESS HAZARDS							
		Asset Integrity Management		Asset integrity threats (over pressure, corrosion etc.)	Injury/fatality from ignited release Environmental pollution from unignited release	H		Carry out HAZOP study for the project facilities
		Excess/zero Level	Closed drain tanks and open drain	Overfill	Could result into environmental pollution	M	See recommendation on HAZOP	

			tank					
2.4	UTILITY SYSTEM HAZARDS							
		Potable water	Utility water	No hazardous scenario		L		
2.5	MAINTENANCE HAZARDS							
		Access Requirements	See maintenance philosophy					
		Heavy Lifting Requirements	See maintenance philosophy					
		Transport		Release of transport hazards during construction	Injury/fatality during construction	H		Develop transportation HSSE plan during construction phase of the project
2.6	CONSTRUCTION/ EXISTING FACILITIES							

		Skid Dimensions (weight handling/equipment (congestion))	Pig traps, chemical injection, containerised packages	Water transport access issues	Project delay	L	Logistic infrastructure	Carry out Logistic Infrastructure Review and Assessment (LIRA)
3	BIOLOGICAL							
		snakes, reptiles		Attack	Injury/fatality from biological hazard release	M	HSSE plans including the induction program	
3.2	CHEMICALS							
		glycol, foam systems, water treatment chemicals	Corrosion inhibitor, Biocides injection and scale inhibitor	Contact with chemicals	Burns or health effects from chemicals	L		Develop chemical handling procedure using SHOC cards, during detailed design
3.3	ERGONOMICS							
		Human machine interface		Stress and ergonomic issues	Health effects	L	Offices provided	

3.4	NOISE							
		Blowdown valves, gas generator	Blow-down valves	Noise from blowdown valves	Hearing defects	M		Develop noise mapping taken into consideration the blowdown valves

Shell Risk Assessment Matrix

SEVERITY	CONSEQUENCES				INCREASING LIKELIHOOD				
	People	Assets	Community	Environment	A	B	C	D	E
					Never heard of in the industry	Heard of in the industry	Has happened in the Organization or more than once in the Industry	Has happened at the Location or more than once per year in the Organization	Has happened more than once per year at the location
0	No injury or health effect	No damage	No impact	No effect					
1	Slight injury or health effect	Slight damage	Slight impact	Slight effect					
2	Minor injury or health effect	Minor damage	Minor impact	Minor effect					
3	Major injury or health effect	Moderate damage	Moderate impact	Moderate effect					
4	PTD or up to 3 fatalities	Major damage	Major impact	Major effect					
5	More than 3 fatalities	Massive damage	Massive impact	Massive effect					